BIOL 230: Epiphyte Activity Lab Report

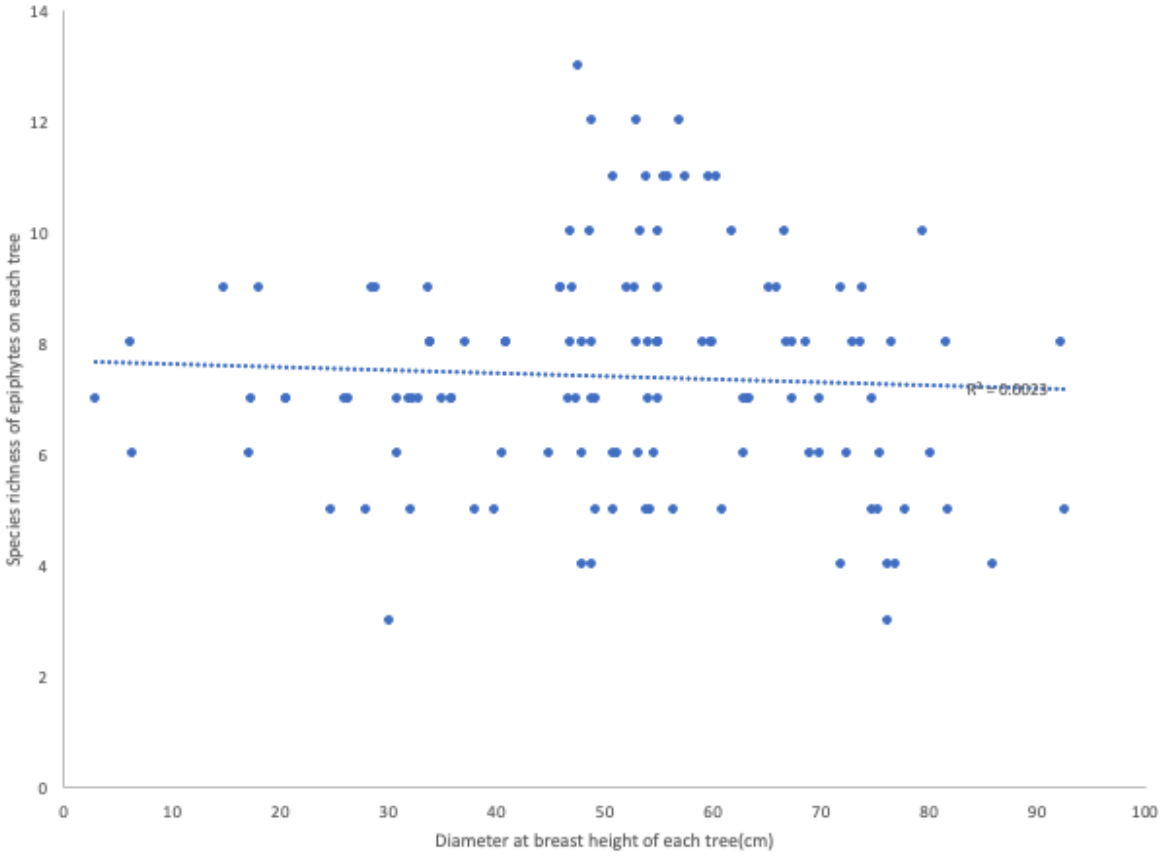


Figure 1. Relationship between tree size and epiphyte species richness. Each point represents one tree (n = 123) surveyed on the UBC Vancouver campus on October 20, 2025. Tree size is measured as diameter at breast height (DBH, cm), and epiphyte species richness represents the number of bryophyte and lichen species observed across 12 quadrat samples per tree. The dashed line indicates the fitted linear regression trendline. Pearson's correlation coefficient was r = –0.10, with a two-tailed P-value = 0.28, and the trend explained very little variation ($R^2 = 0.0023$). Because P > 0.05, the relationship is not statistically significant. Species identification and abundance scoring follow the Epiphyte Lab assignment.

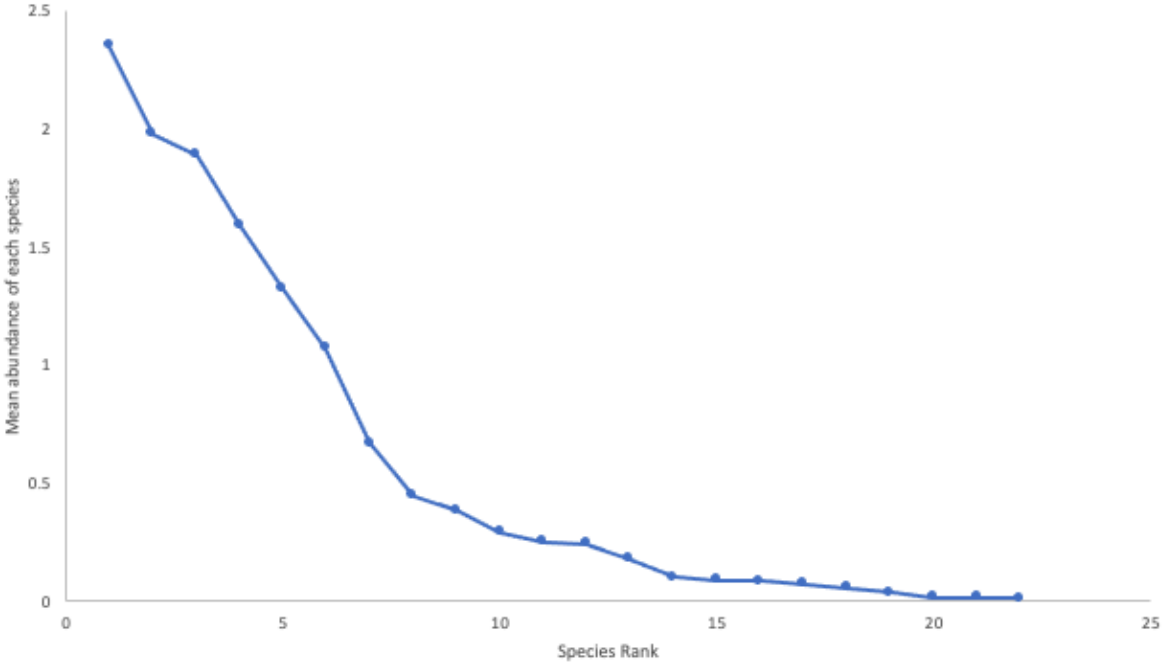


Figure 2. Rank-abundance curve showing the mean abundance of epiphyte species across all sampled trees. Each point represents one epiphyte species (bryophytes and lichens) averaged across all trees in the dataset (n = 123 trees surveyed on the UBC Vancouver campus on October 20, 2025). Species rank is ordered from most to least abundant, and mean abundance reflects the average number of quadrat cells occupied by each species across 12 samples per tree. Points are connected by a line to show the decline in abundance across ranks. Species codes and identification procedures follow the Epiphyte Lab assignment. No trendline is included.

Questions:

1. Tree size can act as a proxy for available habitat area for epiphytes, so both the sampling effect and habitat effect could influence species richness. The sampling effect suggests that larger trees have more surface area, meaning that more individual epiphytes can occur simply due to chance, which increases the likelihood of encountering more species. The habitat effect is also relevant because larger trees provide more microhabitats, such as differences in bark moisture, texture, exposure to light, and vertical height gradients. These microhabitats can support species with different environmental preferences, increasing species richness as tree size increases. Island biogeography theory may apply to a lesser extent, because trees can be considered habitat "islands" within the landscape; larger trees may experience lower local extinction rates than smaller trees due to more stable microclimatic conditions. Therefore, sampling effect, habitat heterogeneity, and, to a more limited degree, island biogeography could all theoretically contribute to a positive relationship between tree size and epiphyte species richness in this system.

2. The scatterplot and correlation show a very weak negative relationship between tree DBH and epiphyte species richness ($r = -0.10$, $R^2 = 0.0023$, $n = 123$) and the result is not statistically significant (two-tailed $P = 0.28$). This indicates that richness does not systematically increase or decrease with tree size in our sample; any apparent pattern is indistinguishable from random variation. Conceptually we might have expected a slight positive relationship because larger trees provide more area and microhabitats, but the observed non-significance suggests other factors (bark chemistry, age vs. size mismatch, shading and moisture differences among sites, management history, or measurement noise) likely overshadow any simple size effect under our sampling.

3. Rank–abundance curves reflect evenness by their slope: a steep initial drop with a long tail indicates low evenness (a few dominant species and many rare ones), whereas a flatter curve indicates higher evenness. Our curve is steep at low ranks and then trails off near zero for many species, so the community appears uneven—dominated by a small number of common epiphytes with many species present at low mean abundance. This pattern is consistent with competitive hierarchies and microhabitat filtering typical of bryophyte and lichen assemblages on tree trunks.